

# Global Superstore Advanced Data Analysis

## *Advanced Data Analysis Document*

### 1. Sales Performance Analysis

---

#### Sales by Segment

The Consumer segment is the leading contributor to sales, generating approximately \$6.51 million, which represents around 51.5% of total sales. This makes Consumer the dominant driver of revenue ahead of the Corporate and Home Office segments.

#### Sales by Region

Central is the strongest-performing region, generating approximately \$2.82 million in sales, the highest of all regions. Africa recorded the lowest regional sales at approximately \$783,773, indicating a substantial performance gap between the two.

#### Sales by Market

APAC is the top-performing market, generating the highest sales of the analysis (approximately \$3.59 million). Canada is the weakest market, with sales of approximately \$66,928 — the lowest of any market in the dataset.

#### Sales Trends

Sales showed an overall increasing trend across the period analysed, although performance fluctuated from month to month rather than growing in a straight line. This is consistent with the profit trend described in Section 7.

### 2. Profitability Analysis

---

#### Profit by Segment

Consumer is also the leading segment by profit, generating approximately \$749,240, or around 51.1% of total profit — closely mirroring its share of sales.

#### Profit by Category

Furniture is the weakest-performing category by profit, recording the lowest profit of the three product categories despite generating meaningful sales volume. This underperformance is examined in detail in Section 6 (Discount Analysis) and Section 8 (Business Problems).

#### Profit by Region

Central generated the highest regional profit, consistent with its position as the top region by sales. Africa recorded the lowest regional profit, again mirroring its position as the weakest region by sales.

#### Profit by Market

APAC produced the highest market profit, while Canada produced the lowest — the same ranking pattern seen in the sales analysis, suggesting Canada's weakness is broad-based rather than isolated to one metric.

### Profit by Sub-Category

Sub-category profit should be reviewed directly in the dashboard's category/sub-category breakdown, with particular attention to sub-categories inside Furniture (e.g. Chairs, Tables, Bookcases), since the category as a whole underperforms on profit.

*Note: insert the specific sub-category profit figures from your Power BI matrix visual here once confirmed.*

## 3. Product Performance Analysis

---

### Top Profitable Products

Identify your top 5–10 products by profit directly from the dashboard's ranked product visual and list them here with their profit values, to contrast against the loss-making products below.

### Loss-Making Products

Several products recorded high sales but negative profit, indicating pricing, cost, or discount issues at the product level:

| Product                          | Sales    | Profit / (Loss) |
|----------------------------------|----------|-----------------|
| Motorola Smart Phone             | \$38,931 | (\$4,447)       |
| Rogers Lockers                   | \$28,214 | (\$2,893)       |
| HowVwer Stove White              | \$32,842 | (\$2,181)       |
| Sacto Executive Leather Armchair | —        | (\$1,833)       |
| Cisco Telepresence System        | —        | (\$1,811)       |

### High Sales vs. Low/Negative Profit

The Motorola Smart Phone is the clearest example of this pattern: it generated \$38,931 in sales — a substantial amount — yet produced a loss of \$4,447. This confirms that sales volume alone is not a reliable indicator of profitability and that product-level review is necessary.

## 4. Geographic Analysis

---

### Regional Performance

Central is the best-performing region overall (highest sales and profit). Africa is the weakest-performing region on both metrics, representing the most significant regional performance gap in the dataset.

### Market Performance

APAC is the best-performing market overall. Canada is the weakest market, with sales of only about \$66,928 — far below the other markets analysed.

### Best and Worst-Performing Locations

| Level  | Best Performer                   | Worst Performer                          |
|--------|----------------------------------|------------------------------------------|
| Region | Central (highest sales & profit) | Africa (~\$783,773 sales, lowest profit) |

| Level  | Best Performer                | Worst Performer                         |
|--------|-------------------------------|-----------------------------------------|
| Market | APAC (highest sales & profit) | Canada (~\$66,928 sales, lowest profit) |

## 5. Customer and Segment Analysis

---

### Sales by Segment

Consumer: ~\$6.51 million (~51.5% of total sales) — the leading segment. Corporate and Home Office contribute the remainder and represent areas for diversification.

### Profit by Segment

Consumer: ~\$749,240 (~51.1% of total profit) — again the leading segment, showing that its profit contribution is proportionate to its sales contribution.

### Top Customers by Sales

Tom Ashbrook is the top customer by sales, at approximately \$40,488.

### Top Customers by Profit

Tamara Chand is the top customer by profit, at approximately \$8,673. Notably, she also ranks second by sales — meaning she is both a high-revenue and a high-profit customer.

### High-Value Customers

Comparing Tom Ashbrook and Tamara Chand shows that the highest-revenue customer is not automatically the highest-profit customer. This means customer value should be assessed using profitability, not sales alone, when prioritising retention and loyalty efforts.

## 6. Discount Analysis

---

### Average Discount by Category

Furniture carries the highest average discount of the three categories, at 17%.

### Discount vs. Sales

Higher discount levels were not consistently associated with higher sales across categories and products — the relationship is mixed rather than linear.

### Discount vs. Profit

Similarly, larger discounts were not consistently associated with higher profitability. In Furniture's case, the highest average discount coincided with the lowest category profit, but discounting is not the only factor at play (see Section 8).

### Categories Negatively Affected by Discounting

Furniture is the category most clearly linked to discount-related profitability pressure, given its combination of the highest average discount and the lowest category profit.

### Products Negatively Affected by Discounting

The loss-making products listed in Section 3 are likely candidates for discount-related margin erosion and should be cross-checked against their individual discount rates in the dashboard to confirm the extent to which discounting contributed to each loss.

## 7. Time-Based Analysis

---

### Monthly Trends

September 2024 was the highest-profit month in the dataset. July 2011 was the lowest-profit month, marking the two extremes of monthly performance.

### Quarterly Performance

Review the quarterly profit and sales visual in the dashboard to identify which quarters consistently outperform, and note this here alongside the monthly extremes above.

### Yearly Performance

Review year-over-year sales and profit totals in the dashboard to confirm the overall growth trend described below at an annual level.

### Best and Worst-Performing Periods

Best: September 2024 (highest profit). Worst: July 2011 (lowest profit).

### Overall Trend

Sales and profit both show an overall increasing trend across the period analysed, with fluctuations across individual months rather than uninterrupted growth.

## 8. Business Problems Investigated

---

### Problem 1: Furniture Profitability — High Shipping and Discount Costs

Furniture is the weakest category by profit. It carries the highest average discount (17%) of any category and also has the highest shipping cost. Because the discount–profit relationship elsewhere in the dataset is mixed rather than consistently negative, Furniture's weak profitability is best explained by the combination of high discounting and high shipping costs, rather than either factor alone.

### Problem 2: Loss-Making Products Despite Strong Sales

Products such as the Motorola Smart Phone, Rogers Lockers, and HowVwer Stove White generate substantial sales but record consistent losses. This points to issues with pricing, product cost, or discount depth at the individual product level that are masked when looking only at category or segment totals.

### Problem 3: Geographic Underperformance — Africa and Canada

Africa is the weakest-performing region (lowest sales and profit) and Canada is the weakest-performing market (lowest sales and profit). Both represent significant gaps relative to top performers (Central and APAC) and warrant deeper investigation into demand, pricing, and product mix before further investment decisions are made.

### Problem 4: Customer Value — Sales Leadership vs. Profit Leadership

Tom Ashbrook leads by sales, but Tamara Chand leads by profit while ranking only second in sales. This shows that the business's highest-revenue customer is not its most valuable customer by profit, which has implications for how the company prioritises retention and loyalty investment.